

Gel Electrophoresis

This protocol outlines the proper setup and use of agarose gel electrophoresis to separate DNA fragments based on molecular weight under an electrical current. By running a DNA ladder alongside samples, the size of DNA fragments can be estimated and compared. A 1% agarose gel is used as a standard condition for effective separation. Gel electrophoresis is commonly applied to verify PCR products, confirm plasmid inserts, assess DNA digestion, and roughly estimate DNA concentration based on band intensity.

(casting the gel takes about 40min)

(loading and running the gel takes about 30min)

Reagents

✓	Name	Quantity	Location/Notes
☐	Agarose Powder (NOT Agar Powder)	0.5g	
☐	1x TAE (Tris-Acetate-EDTA) buffer	50 mL	10× TAE 1:10 with ddH ₂ O to obtain 1× TAE
☐	DyeNA View DNA gel stain	5 uL	
☐	6x Purple DNA loading dye	1 uL	Per sample
☐	1 kb DNA ladder	3 uL	
☐	Sample	5uL	(each)

Equipment

✓	Name	Quantity	Location/Notes
☐	Conical flask	1	
☐	Microwave	1	
☐	Gel tray	1	
☐	Gel caster	1	
☐	8-well comb	1	
☐	Microcentrifuge	1	
☐	Gel Electrophoresis Unit	1	
☐	Chemidoc	1	
☐	1.5mL Microcentrifuge tubes / PCR tubes	Samples	
☐	Weigh boat	1	

PPE:

- Gloves
- Goggles
- Thermal gloves

Instructions

[Preparing the 1% agarose gel]

Note: The quantity of gel prepared can be adjusted based on the size of the casting tray. Be sure to adjust the amount of agarose and gel stain as well.

Gel Tray Dimensions (cm)	Agarose Gel Volume Range
7 x 8 cm	40 ml
9 x 11 cm	70 – 80 ml
12 x 14 cm	120 – 130 ml

1. In a conical flask, mix 0.5 g of agarose powder and 50mL of TAE buffer.
2. Swirl the agarose solution to mix it well.
3. Heat the agarose solution in a microwave until it just boils (30 sec – 1.5 min).
4. Take out the conical flask from the microwave and swirl the conical flask until the agarose solution is clear and without undissolved agarose powder or lumps.

Note: Careful after step 4 as the flask may be hot, wear thermal gloves if need be to prevent burns. (or hold with paper towel)

5. Wait for the solution to cool until it can be handled comfortably (50°C). Vortex the [DyeNA-View](#) tube for 2–3 seconds. Add 5uL of DyeNA-View Stain into the agarose solution and swirl the conical flask until the colour becomes uniform.

[Casting the Gel]

1. Secure a gel tray tightly onto a gel caster and place an 8-well comb or 15-well comb into the tray.
Alternatively, you can also put in combs immediately after pouring
2. Carefully pour the agarose solution into the tray

Note: Ensure that there are no bubbles by pouring slowly. Or move them using a pipette tip to the edges where they will not affect visualization.

3. Cool down the solution for at least 30 minutes to solidify the agarose gel.

[Preparing the samples]

Note: Adjust the amount of dye based on the volume of your sample. 5 μ L sample is sufficient for most minipreps and PCRs

Note: Loading dye is 6x so add loading dye at a 5:1 ratio (sample:dye), e.g., 5 μ L sample + 1 μ L dye

1. Pipet 5ul of sample into a new microcentrifuge or PCR tube.
2. Add 1ul of 6X DNA loading dye to the DNA samples. Pipette up and down until the solution is homogenous.
3. You can spin down tubes to get solution to bottom if needed

Note: if needed, samples with loading dye can be stored in the -20 for later use

[Setting up the gel]

1. After the agarose gel has solidified in the gel tray, remove the gel tray from the gel caster and place it in a buffer tank with pre-filled 1x TAE buffer. IMPORTANT: the gel must be fully submerged in the buffer.

Note: you can use a scalpel or razor to cut the gel to a smaller size to avoid wasting gel

2. Add 3 ul DNA ladder and 6 ul mixture of sample + dye into the wells. Ensure to note down what is in each lane.

Note: You can use a [1 kb](#) or [100 bp](#) ladder depending on the expected size of your sample(s).

Note: Avoid pipetting to the second stop to avoid producing bubbles in the well and disrupting the sample.

3. Close the buffer tank with the tank lid, ensuring that the colour of the electrodes on the tank lid is the same as the one on the buffer tank.
4. Run the gel electrophoresis at 100V for up to 60 minutes, making sure that bubbling occurs when the gel electrophoresis begins. The time will depend on the distance travelled by the samples, make sure they do not run off.

Note: You can adjust the time based on the size of bands you are expecting. Run for longer if you are expecting larger bands (>1 kb) to allow the ladder to separate enough. If after visualization bands are not separated enough, you can place it back in the tank and run for longer.

Note: Keep an eye on the gel to make sure samples do not run off and the gel remains straight in the buffer (not tilted)

5. Upon finishing gel electrophoresis, remove gel from apparatus and visualize using the chemidoc using UV light.
6. Save the gel image.
7. Wrap any extra gel in plastic wrap tightly and store in the fridge. Any gels left open to air will quickly dry out

Notes

N/A

References

- As mentioned in lab meeting, largely based upon last year's protocol